

2. (a) The diagram shows the colours of universal indicator at different pH values.

Colour	red	orange	yellow	green	blue	navy blue	purple
pH	1 - 2	3 - 4	5 - 6	7	8 - 9	10 - 11	12 - 13

- (i) Sodium hydroxide solution turns universal indicator purple.

Put a **tick** (✓) in the box next to the correct statement.

[1]

sodium hydroxide is a strong alkali

☐

sodium hydroxide is a weak alkali

☐

sodium hydroxide is a strong acid

☐

sodium hydroxide is a weak acid

☐

- (ii) Ethanol is neutral. State the **colour** universal indicator would turn in ethanol.

[1]

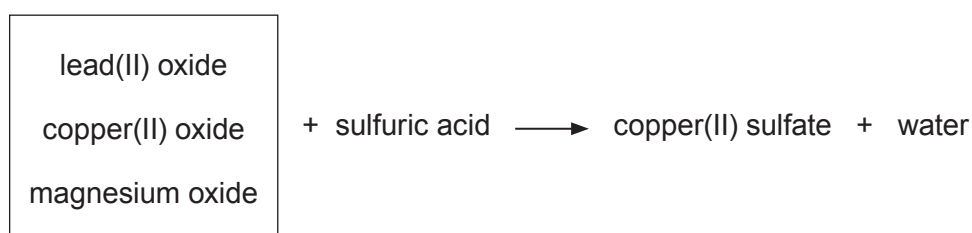
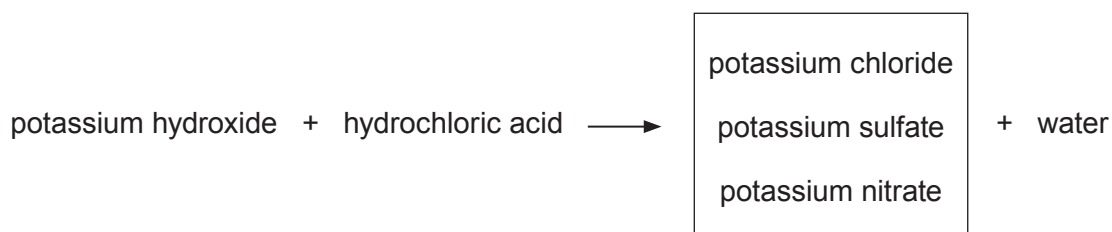
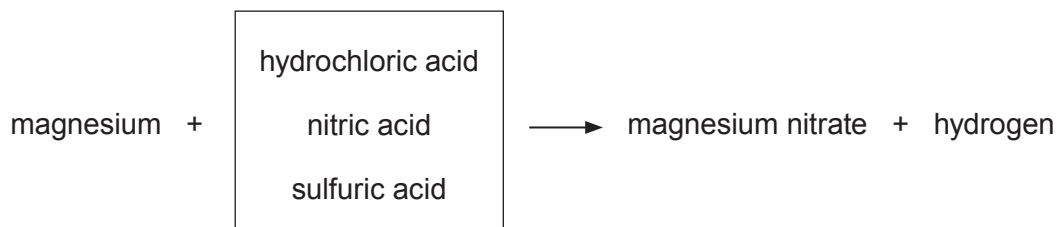
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(b) The word equations show reactions of acids with a metal, an alkali and a base.

Circle the correct substance in the box to complete each equation.

[3]



(c) A student was asked how he would prepare a sample of zinc chloride crystals from zinc.

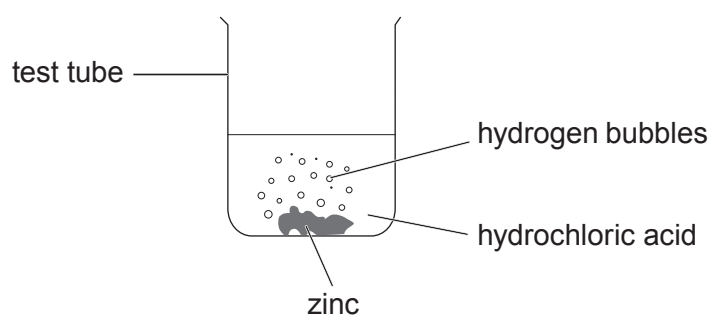
The word equation for the reaction is



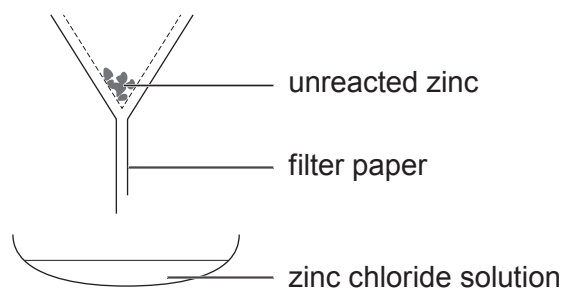
The diagrams below were taken from the student's work book.

There is **one** error in **each** of the student's diagrams.

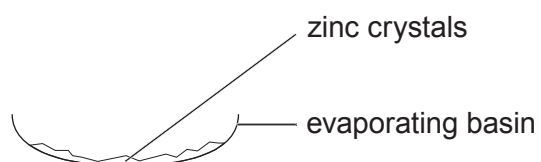
Step 1



Step 2



Step 3



(i) **Circle** the error in each diagram. [3]

(ii) Zinc chloride contains the ions Zn^{2+} and Cl^- .

Underline the correct formula of zinc chloride. [1]



(iii) Put a **tick** (✓) in the box next to the test to show that the gas given off is hydrogen. [1]

the gas pops with a lighted splint

☐

the gas turns limewater milky

☐

the gas relights a glowing splint

☐